Complete formula reference

Mechatronics: slides + homework 01–06

This reference combines the formulas in all 438 pages of *Slides.pdf* with the relationships needed for calculation-based questions in the 32-page *HW_fall_2026.pdf*. Repeated equations and algebraic rearrangements are grouped into one entry. Each entry names the relationship and defines its symbols.

How to read the sources. All page numbers refer to the PDF page count, beginning with page 1, not the printed slide or Blackboard page number. "Derived" means a formula has been written from a described relationship, circuit, or homework requirement. Concept-only homework questions are omitted from the lookup table. Closely related recognition questions are included when an equation directly describes the requested behavior.

Notation. Symbols are defined locally because the slides reuse letters. Δ means a finite change; d/dt means a time derivative; $\int$ means an integral; $\sum$ means a sum. Unless otherwise stated, ln is the natural logarithm and $\log_{10}$ is base 10. $\pi \approx 3.14159$ is the circle constant; a dot over a quantity indicates its rate of change with time. Match units before substituting. Gain ratios, strain and duty-cycle fractions are dimensionless. $\mu = 10^{-6}$, m $= 10^{-3}$ as a unit prefix, k $= 10^{3}$, and M $= 10^{6}$.

Corrections are visible. A few source equations contain sign, approximation, labeling, or unit inconsistencies. The usable equation is given in the main entry, with the discrepancy identified beside it. This is a formula reference, not a set of worked homework solutions.

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1 Control and measurement performance

F01 Negative-feedback error signal

Slides p. 40; HW01 Q1 (p. 1)

$$e(t) = r(t) - y_m(t)$$

Terms: e: control error; r: reference or set-point signal; y_m: measured feedback signal; t: time. All three signals must use compatible scales and units.
Use: Subtract the measured value from the desired value. This sign convention differs from measurement error in F03.

F02 Measurement span

Slides pp. 159, 247

$$S_x = x_{\max} - x_{\min}$$

Terms: S_x: input span; $x_{\max}$ and $x_{\min}$: upper and lower limits of the measurable input. The *range* is the interval $[x_{\min}, x_{\max}]$; the *span* is its width. For an output span, use $S_y = y_{\max} - y_{\min}$.

F03 Signed measurement error

Slides p. 160; HW03 Q10 (p. 16)

$$e_m = x_m - x_{\text{true}}$$

Terms: e_m: measurement error; x_m: indicated or measured value; x_{true}: true value. The magnitude of the error is $|e_m|$. A low reading gives a negative error.

F04 Full-scale percentage error

Slides pp. 161, 164, 171; HW03 Q5 (pp. 14–15)

$$E_{\max} = \frac{p}{100} S, \qquad e \in [-E_{\max}, E_{\max}]$$

Terms: p: stated error percentage, entered as a number such as 0.15 for 0.15%; S: full-scale input or output span on the same scale as the requested error; $E_{\max}$: maximum error magnitude; e: possible signed error.
Use: An error stated as a percentage of full scale is based on the span, not on the current reading. For percentage-of-reading specifications, replace S by the reading's magnitude.

F05 Worst-case combination of error limits

Slides pp. 161, 249 (derived from the practice problem)

$$E_{\text{worst}} = \sum_{i=1}^{m} E_i \quad \text{or} \quad E_{\text{worst}} = \frac{S}{100} \sum_{i=1}^{m} p_i$$

Terms: E_i: magnitude of error limit i; p_i: corresponding percentage of the same full-scale span S; m: number of error contributions; E_{worst}: combined worst-case bound. This is a sum of bounds, not a statistical root-sum-square uncertainty.

F06 Sensor sensitivity and linear calibration

Slides pp. 162, 164, 250; relationship written in general form

$$K = \frac{\Delta y}{\Delta x}, \qquad y - y_0 = K(x - x_0)$$

Terms: K: sensitivity, with units of output per input; x, y: input and output; x_0, y_0: reference input and output; $\Delta x, \Delta y$: corresponding changes. The form $K = y/x$ applies when the calibration passes through the origin.

F07 Endpoint straight line and nonlinearity error

Slides pp. 164, 250 (derived calculation form)

$$y_{\text{line}}(x) = y_{\min} + \frac{y_{\max} - y_{\min}}{x_{\max} - x_{\min}}(x - x_{\min})$$

$$e_{\text{NL}}(x) = y(x) - y_{\text{line}}(x), \qquad p_{\text{NL}} = 100\frac{\max_x |e_{\text{NL}}(x)|}{S_y}$$

Terms: y_{line}: ideal straight-line output; $(x_{\min}, y_{\min})$ and $(x_{\max}, y_{\max})$: endpoint calibration pairs; $y(x)$: actual output; e_{NL}: deviation from that line; S_y: full output span; p_{NL}: maximum nonlinearity as a percentage of full scale. For a single specified input, omit the maximum and evaluate the absolute deviation at that input.

F08 Hysteresis error

Slides p. 163; mathematical form of the definition

$$H = \max_x |y_{\uparrow}(x) - y_{\downarrow}(x)|, \qquad p_H = 100\frac{H}{S_y}$$

Terms: H: maximum output difference at the same input; $y_{\uparrow}, y_{\downarrow}$: outputs obtained while input increases or decreases; x: input; S_y: full output span; p_H: full-scale percentage hysteresis.

F09 Repeatability error

Slides p. 165

$$p_{\text{rep}} = 100\frac{y_{\max,\text{rep}} - y_{\min,\text{rep}}}{S_y}$$

Terms: p_{rep}: repeatability error as a full-scale percentage; $y_{\max,\text{rep}}$ and $y_{\min,\text{rep}}$: largest and smallest outputs from repeated measurements of the *same* input; S_y: full output span.

F10 Drift and temperature-induced output change

Slides pp. 162, 166–167, 171; definitions expressed mathematically

$$\Delta y_{\text{drift}} = y(t_2) - y(t_1), \qquad p_{\text{drift}} = 100\frac{\Delta y_{\text{drift}}}{S_y}$$

$$\Delta y_T = K_T \Delta T$$

Terms: $y(t_1), y(t_2)$: outputs at two times for a constant input; Δy_{drift}: signed drift; p_{drift}: drift as a percentage of output span S_y; K_T: output sensitivity to temperature; ΔT: temperature change; Δy_T: resulting output shift. If the specification is p_T% full scale per degree, use $K_T = (p_T/100)S_y$ per degree. Use absolute values when an error magnitude is requested.

F11 First-order sensor step response and time constant

Slides pp. 174–177; HW04 Q9 (p. 21); general response derived from the time-constant model

$$y(t) = y_\infty + (y_0 - y_\infty)e^{-t/\tau}$$

$$\frac{y(\tau) - y_0}{y_\infty - y_0} = 1 - e^{-1} \approx 0.6321, \qquad \frac{y(\tau) - y_\infty}{y_0 - y_\infty} = e^{-1} \approx 0.3679$$

Terms: $y(t)$: sensor output at time t after a step; y_0: initial output; y_∞: final steady-state output; τ: time constant; e: Euler's number, approximately 2.71828.

Source note: The percentages describe the fraction of the *total change* completed or remaining at $t = \tau$. A time constant is not itself a percentage, and is not "63.2% of the response time" as stated in the summary on p. 248.

F12 First-order response, rise and settling times

Slides pp. 174–177; derived from F11 for a monotonic first-order response

$$t_p = -\tau \ln(1 - p), \qquad t_{10\text{–}90} = \tau \ln 9 \approx 2.197\tau, \qquad t_s = -\tau \ln \epsilon_s$$

Terms: t_p: time to complete fraction p of the total step change, where $0 < p < 1$; $t_{10\text{–}90}$: time from 10% to 90% response; t_s: time to enter a final-value band; ϵ_s: allowed remaining-error fraction of the total change, such as 0.02 for 2%; τ: time constant. These formulas assume no dead time and a first-order model.

2 Displacement, strain and motion sensors

F13 Unloaded potentiometer / voltage-divider relationship

Slides p. 184

$$\frac{V_o}{V_s} = \frac{R_{23}}{R_{13}} = a, \qquad a = \frac{x}{L} \text{ or } \frac{\theta}{\theta_{\max}}$$

Terms: V_s: voltage across the complete track; V_o: output voltage from slider to reference terminal; R_{23}: slider-to-reference resistance; R_{13}: total track resistance; a: fractional slider position; x: linear displacement over track length L; θ: angular displacement over total usable angle $\theta_{\max}$. Assumes a uniform track and negligible output loading.

F14 Loaded potentiometer

Slides p. 185; derived from the shown circuit

$$R_p = \frac{aR_T R_L}{aR_T + R_L}, \qquad V_L = V_s \frac{R_p}{(1-a)R_T + R_p} = V_s \frac{a}{1 + a(1-a)R_T/R_L}$$

Terms: R_T: total track resistance; R_L: resistance connected across the output; a: slider fraction measured from the reference end; R_p: parallel combination of the lower track section aR_T and R_L; V_s: supply voltage; V_L: loaded output. Finite R_L makes the position-to-voltage relation nonlinear.

F15 Strain and strain-gauge factor

Slides pp. 186–188; HW03 Q6 (p. 15)

$$\varepsilon = \frac{\Delta L}{L_0}, \qquad \frac{\Delta R}{R_0} = G_F \varepsilon, \qquad \varepsilon = \frac{R - R_0}{G_F R_0}$$

Terms: ε: axial engineering strain; L_0: original length; ΔL: change in length; R_0: unstrained resistance; R: strained resistance; $\Delta R = R - R_0$: resistance change; G_F: gauge factor, called G in the slides. Strain is dimensionless; $1\,\mu\varepsilon = 10^{-6}$ strain. Assumes temperature effects have been removed or are negligible.

F16 Parallel-plate capacitance

Slides p. 190

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

Terms: C: capacitance (F); ε_r: relative permittivity of the dielectric; ε_0: vacuum permittivity (F/m); A: overlapping plate area (m^2); d: plate separation (m). Assumes a uniform dielectric and negligible fringing.

F17 LVDT differential output

Slides pp. 192–193; relationship expressed from the description

$$V_o = V_{s1} - V_{s2} \approx K_x x$$

Terms: V_o: differential output with the chosen secondary winding polarity; V_{s1}, V_{s2}: secondary-coil voltages; x: core displacement from the null position; K_x: calibrated displacement sensitivity (V per unit length). The signed linear relation applies near the null; AC phase or demodulation conveys the displacement direction.

F18 Encoder angular resolution and displacement

Slides pp. 198–199; HW03 Q7 (p. 15)

$$\Delta\theta = \frac{360^\circ}{N_c}, \qquad \theta = \frac{360^\circ N_p}{N_c}$$

Terms: $\Delta\theta$: angular increment per count; N_c: effective counts per revolution; N_p: accumulated counts; θ: rotation from the chosen starting point. For one count per slot, N_c is the number of slots. For an n-bit absolute encoder, $N_c = 2^n$, with n the number of binary tracks/bits. Use effective counts, including any specified quadrature multiplication.

F19 Tracking-wheel displacement and encoder pulses

HW03 Q9 (p. 16); derived using wheel circumference

$$C_w = 2\pi r, \qquad N_p = \frac{N_c x}{2\pi r}, \qquad x = \frac{2\pi r N_p}{N_c}$$

Terms: C_w: wheel circumference; r: wheel radius; x: linear travel; N_c: encoder pulses/counts per revolution; N_p: pulse count corresponding to that travel. Use the same length units for r and x, and assume no wheel slip.
Source note: The continuous calculation can give a fractional count. Whole completed counts depend on the starting phase; the HW03 Q9 key is consistent with truncation to completed pulses rather than rounding to the nearest integer.

F20 Pulse-based speed and tachogenerator output

Slides pp. 207–209; relationships expressed from the descriptions

$$f_p = \frac{\Delta N_p}{\Delta t}, \qquad n_{\text{rpm}} = \frac{60 f_p}{N_c}, \qquad \omega = \frac{2\pi f_p}{N_c}, \qquad E = K_\omega \omega$$

Terms: f_p: pulse frequency (counts/s); ΔN_p: counts in interval Δt (s); N_c: counts per revolution; n_{rpm}: shaft speed in revolutions/min; ω: angular speed (rad/s); E: calibrated tachogenerator voltage amplitude or rectified output; K_ω: tachogenerator sensitivity (V per rad/s). The last relation is the linear amplitude model described for the generator.

F21 Hall-effect voltage

Slides p. 204

$$V_H = K_H \frac{BI}{t_p}$$

Terms: V_H: transverse Hall voltage (V); K_H: Hall coefficient (m^3/C); B: perpendicular magnetic flux density (T); I: current through the plate (A); t_p: plate thickness (m), called t in the slides. Sign depends on the charge carriers and chosen polarity.

3 Pressure, flow, level and temperature

F22 Gauge and differential pressure

Slides p. 215; mathematical forms of the definitions

$$p_g = p_{\text{abs}} - p_{\text{atm}}, \qquad \Delta p = p_1 - p_2$$

Terms: p_g: gauge pressure; p_{abs}: pressure measured relative to a vacuum; p_{atm}: atmospheric pressure; p_1, p_2: pressures at the two measurement ports; Δp: differential pressure. All pressures use the same units.

F23 Bernoulli equation for a horizontal flow passage

Slides p. 221

$$\frac{v_1^2}{2g} + \frac{p_1}{\rho g} = \frac{v_2^2}{2g} + \frac{p_2}{\rho g}$$

Terms: v_1, v_2: upstream and constriction velocities (m/s); p_1, p_2: corresponding pressures (Pa); ρ: fluid density (kg/m^3); g: gravitational acceleration (m/s^2). Applies to steady, incompressible flow along a streamline with negligible losses and equal elevations. Each term is a head (m).

F24 Continuity, volumetric flow and mass flow

Slides p. 222

$$Q = A_1 v_1 = A_2 v_2, \qquad \dot{m} = \rho Q$$

Terms: Q: volumetric flow rate (m^3/s); A_1, A_2: flow cross-sectional areas (m^2); v_1, v_2: average velocities (m/s); $\dot{m}$: mass flow rate (kg/s); ρ: constant fluid density (kg/m^3). Assumes steady incompressible flow with no leakage.

F25 Ideal constriction / orifice flow relationship

Slides pp. 222–223

$$Q = \frac{A_2}{\sqrt{1-(A_2/A_1)^2}}\sqrt{\frac{2(p_1-p_2)}{\rho}}$$

Terms: Q: ideal volumetric flow rate; A_1: upstream area; A_2: constriction area; $p_1 - p_2$: pressure drop; ρ: density. Use SI units to obtain m^3/s.
Source note: The numerator on p. 222 is labeled only A; it must be the constriction area A_2 to follow from F23–F24. A real orifice requires a calibrated discharge coefficient multiplying the ideal flow.

F26 Turbine flowmeter calibration

Slides p. 224; proportionality expressed mathematically

$$f_p = K_Q Q \qquad \Longrightarrow \qquad Q = \frac{f_p}{K_Q}$$

Terms: f_p: measured turbine pulse frequency (pulses/s); Q: volumetric flow rate; K_Q: calibrated meter factor (pulses per unit volume). This represents the approximate linear relation described in the slide.

F27 Tank volume, liquid weight and load-cell sizing

Slides pp. 225, 243

$$A = \frac{\pi D^2}{4}, \qquad V = Ah, \qquad W_\ell = \rho g A h$$

$$W_{\text{total}} = W_0 + W_\ell, \qquad F_{\text{cell}} = \frac{W_{\text{total}}}{n_c}, \qquad \Delta F_{\text{cell}} = \frac{\rho g A \Delta h}{n_c}$$

Terms: A: constant tank cross-sectional area; D: internal diameter for a circular tank; V: liquid volume; h: liquid height; ρ: density; g: gravitational acceleration; W_ℓ: liquid weight (N); W_0: empty-tank weight (N); W_{total}: combined weight; n_c: number of equally loaded cells; F_{cell}: force per cell; Δh: smallest level change to detect; ΔF_{cell}: corresponding required force increment per cell. If the tank quantity is given in kilograms, it is mass: multiply by g to obtain weight.

F28 Hydrostatic liquid-level measurement

Slides pp. 225, 227

$$\Delta p = p_{\text{bottom}} - p_{\text{surface}} = \rho g h, \qquad h = \frac{\Delta p}{\rho g}$$

Terms: p_{bottom}: pressure at the submerged measurement point; p_{surface}: gas pressure above the liquid; Δp: pressure caused by the liquid column; ρ: liquid density; g: gravitational acceleration; h: vertical liquid depth above the measurement point.

F29 Linear resistance-temperature relationship / RTD

Slides pp. 229, 315, 327

$$R(T) = R_0[1 + \alpha(T - T_0)], \qquad \Delta R = R_0 \alpha \Delta T$$

Terms: $R(T)$: resistance at temperature T; R_0: resistance at reference temperature T_0; α: temperature coefficient of resistance (K^{-1} or $^\circ\mathrm{C}^{-1}$ for temperature differences); $\Delta T = T - T_0$; $\Delta R = R(T) - R_0$. The slides take $T_0 = 0\,^\circ\mathrm{C}$, giving $R_t = R_0(1 + \alpha t)$. Valid over the range where the linear approximation applies.

F30 Thermodiode / Shockley diode relationship

Slides p. 231

$$I = I_0\left[\exp\left(\frac{q_e V}{k_B T}\right) - 1\right], \qquad V = \frac{k_B T}{q_e}\ln\left(1 + \frac{I}{I_0}\right)$$

Terms: I: diode current; I_0: reverse saturation current; V: junction voltage; T: absolute temperature (K); q_e: positive elementary-charge magnitude (called e in the slide); k_B: Boltzmann constant (called k); exp: exponential function. This is the slide's ideality-factor-one model.
Source note: I_0 is temperature-dependent in a real diode. Treating it as constant yields the simplified proportionality described by the slide; it is not a general calibrated temperature law.

F31 Thermocouple law of intermediate temperatures

Slides pp. 235–237, 325–326

$$E(T_h, 0) = E(T_h, T_c) + E(T_c, 0)$$

Terms: $E(T_h, T_c)$: measured thermocouple EMF (voltage) between hot-junction temperature T_h and cold-junction temperature T_c; $E(T_c, 0)$: cold-junction compensation EMF from a table referenced to $0\,^\circ$C; $E(T_h, 0)$: equivalent EMF with a $0\,^\circ$C reference. All EMFs must refer to the same thermocouple type. EMF means *electromotive force*.

F32 Local linear thermocouple approximation

Slides p. 328

$$E \approx k_{\mathrm{TC}}(T_h - T_c)$$

Terms: E: thermocouple EMF; k_{TC}: approximately constant thermocouple sensitivity (V/K, often μV/$^\circ$C); T_h, T_c: hot- and cold-junction temperatures. The slide's $e = kt$ is this relation with a zero-degree reference and a small temperature interval. Use F31 with tables when nonlinearity matters.

4 Circuit fundamentals and amplifier formulas

F33 Ohm's law

Used in Slides pp. 212, 253–257, 311–312, 342

$$V = IR$$

Terms: V: voltage drop (V); I: current (A); R: resistance (Ω). This is the supporting resistor relationship used throughout the circuits. For sinusoidal phasors, the analogous impedance relation is $\widetilde{V} = \widetilde{I}Z$, where Z is complex impedance (Ω).

F34 Series and parallel resistance

Slides pp. 185, 255–257; circuit relationships expressed explicitly

$$R_s = \sum_i R_i, \qquad \frac{1}{R_p} = \sum_i \frac{1}{R_i}, \qquad R_a \parallel R_b = \frac{R_a R_b}{R_a + R_b}$$

Terms: R_s: equivalent series resistance; R_p: equivalent parallel resistance; R_i: individual resistance; $R_a \parallel R_b$: parallel combination of two resistors. Series elements carry the same current; parallel elements have the same voltage.

F35 Voltage loading in a sensor-amplifier-display chain

Slides p. 257

$$V_{\mathrm{in}} = V_s \frac{R_{\mathrm{in}}}{R_s + R_{\mathrm{in}}}$$

$$V_d = G V_{\mathrm{in}} \frac{R_d}{R_{\mathrm{out}} + R_d} = G V_s \frac{R_{\mathrm{in}}}{R_s + R_{\mathrm{in}}} \frac{R_d}{R_{\mathrm{out}} + R_d}$$

Terms: V_s: unloaded sensor voltage; R_s: sensor output resistance; R_{in}: amplifier input resistance; V_{in}: actual amplifier input voltage; G: amplifier's unloaded voltage gain; R_{out}: amplifier output resistance; R_d: display/load resistance; V_d: voltage delivered to the display.

Source note: Negligible loading requires $R_{\mathrm{in}} \gg R_s$ and $R_d \gg R_{\mathrm{out}}$. The inequalities printed at the bottom of p. 257 are reversed.

F36 Voltage gain and port impedance

Slides pp. 262, 271

$$G = \frac{V_o}{V_i}, \qquad V_o = GV_i, \qquad Z_{\text{in}} = \frac{\widetilde{V}_i}{\widetilde{I}_i}$$

$$Z_{\text{out}} = -\frac{\Delta\widetilde{V}_o}{\Delta\widetilde{I}_L} \quad \text{at fixed input}$$

Terms: G: voltage gain; V_i, V_o: input and output voltages; Z_{in}: input impedance; $\widetilde{V}_i, \widetilde{I}_i$: input voltage and current phasors; Z_{out}: output/source impedance; $\Delta\widetilde{V}_o$: output-voltage change caused by a load-current change $\Delta\widetilde{I}_L$; current I_L is positive out of the amplifier.
Source note: The slide's simple output-voltage/output-current wording should be understood as a port/test relationship. The operating ratio V_o/I_L is the load impedance, not generally the amplifier's output impedance.

F37 Ideal op-amp relationships

Slides pp. 267–269; also useful for HW04 Q2 (pp. 17–18)

$$V_o = A_{\text{OL}}(V_+ - V_-), \qquad A_{\text{OL}} \to \infty, \quad Z_{\text{in}} \to \infty, \quad Z_{\text{out}} = 0$$

$$I_+ = I_- = 0, \qquad V_+ \approx V_- \quad \text{with linear negative feedback}$$

Terms: V_o: output voltage; A_{OL}: open-loop differential gain; V_+, V_-: non-inverting and inverting *input* voltages; I_+, I_-: input currents; $Z_{\text{in}}, Z_{\text{out}}$: input and output impedances.
Use: Apply the equal-input-voltage approximation only when negative feedback keeps the amplifier in its linear region. It does not apply to a saturated comparator.

F38 Non-inverting amplifier

Slides p. 272

$$G = 1 + \frac{R_f}{R_g}, \qquad V_o = \left(1 + \frac{R_f}{R_g}\right) V_i$$

Terms: G: closed-loop gain; V_i: input at the non-inverting terminal; V_o: output; R_f: feedback resistor from output to the inverting terminal (R_2 in the slide); R_g: resistor from the inverting terminal to ground (R_1). Assumes an ideal op-amp operating within its output range.

F39 Voltage follower / unity-gain buffer

Slides pp. 273–275; HW04 Q5 (pp. 18–19)

$$V_o = V_i, \qquad G = 1$$

Terms: V_i: voltage applied to the non-inverting input; V_o: output directly connected to the inverting input; G: voltage gain. The buffer ideally reproduces voltage while reducing loading on the source.

F40 Inverting amplifier

Slides pp. 276–277; HW04 Q10 (pp. 21–22)

$$G = -\frac{R_f}{R_i}, \qquad V_o = -\frac{R_f}{R_i} V_i$$

Terms: V_i: input voltage; V_o: output; R_i: resistor from input signal to the inverting terminal (R_1); R_f: feedback resistor (R_2); G: signed voltage gain. The non-inverting input is grounded. The negative sign denotes inversion.

F41 Inverting summing amplifier

Slides p. 278

$$V_o = -R_f \sum_{j=1}^{m} \frac{V_j}{R_j}$$

$$R_j = R_i \text{ for all } j \quad \Longrightarrow \quad V_o = -\frac{R_f}{R_i} \sum_{j=1}^{m} V_j$$

Terms: V_o: output voltage; V_j: input voltage j; R_j: resistor for that input; R_f: feedback resistance; m: number of inputs; R_i: common input-resistance value in the equal-resistor case. In the slide, inputs are V_A, V_B, V_C, input resistors are R_A, R_B, R_C, and $R_f = R_2$.

F42 Difference / subtractor amplifier

Slides p. 279; HW04 Q6 (pp. 19–20)

$$V_o = k(V_2 - V_1), \qquad k = \frac{R_f}{R_i} = \frac{R_b}{R_a}$$

Terms: V_1: signal connected through R_i to the inverting input; V_2: signal connected through R_a to the non-inverting input; R_f: feedback resistor; R_b: resistor from the non-inverting input to ground; k: matched resistance ratio; V_o: output. In the slide, $R_i = R_a = R_1$ and $R_f = R_b = R_2$.
Use: HW04 Q6 uses this circuit with a matched ratio of 3. It is a difference amplifier even though both inputs appear on the left of the symbol.

F43 Inverting integrating amplifier

Slides p. 280

$$V_o(t_2) - V_o(t_1) = -\frac{1}{RC} \int_{t_1}^{t_2} V_i(t)\, dt$$

Terms: $V_o(t_1), V_o(t_2)$: output at initial and final times; $V_i(t)$: input waveform; R: input resistance; C: feedback capacitance; t_1, t_2: integration limits. The integral has units V·s and RC has units s. Applies to the ideal integrating circuit before saturation.

F44 Inverting differentiating amplifier

Slides p. 281

$$V_o = -RC \frac{dV_i}{dt}$$

Terms: V_o: output voltage; V_i: input voltage; dV_i/dt: input slope; R: feedback resistance; C: input capacitance; t: time. This is the ideal differentiator model; the series input resistor shown in the practical circuit limits its frequency range.

F45 Three-op-amp instrumentation amplifier

Slides p. 283; sign corrected to match the drawn circuit

$$V_{o1} - V_{o2} = \left(1 + \frac{2R_2}{R_1}\right)(V_{i1} - V_{i2})$$

$$V_o = \frac{R_5}{R_4}\left(1 + \frac{2R_2}{R_1}\right)(V_{i2} - V_{i1})$$

Terms: V_{i1}, V_{i2}: upper and lower input voltages; V_{o1}, V_{o2}: first-stage upper and lower outputs; R_1: resistor linking the first-stage inverting nodes; $R_2 = R_3$: equal first-stage feedback resistors; R_4, R_5: inverting-path input and feedback resistors of the final subtractor; R_6, R_7: corresponding non-inverting divider resistors; V_o: final output. The second relation requires $R_5/R_4 = R_7/R_6$.
Source note: The printed first-stage equation reverses the output difference without reversing the input difference. The first equation above corrects that sign; the second follows from the final subtractor shown.

F46 Logarithmic amplifier

Slides p. 285; normalized diode-based form supplied to clarify the printed shorthand

$$V_o = -n_d V_T \ln\left(1 + \frac{V_i}{RI_S}\right) \approx -n_d V_T \ln\left(\frac{V_i}{RI_S}\right), \qquad V_T = \frac{k_B T}{q_e}$$

Terms: V_o: output voltage; V_i: positive input voltage for this polarity; R: input resistor; I_S: feedback-diode saturation current; n_d: diode ideality factor (1 in the ideal model); V_T: thermal voltage; k_B: Boltzmann constant; T: absolute temperature; q_e: elementary charge. The approximation assumes $V_i/(RI_S) \gg 1$.
Source note: The slide prints $-C\ln(V_i/R) = K\ln V_i$, with C, K unspecified. This omits a reference current/voltage and an additive constant. A logarithm must act on a dimensionless ratio. Here RI_S provides the voltage reference; the slide's C is a coefficient, not the capacitor in F43–F44.

F47 Comparator output

Slides p. 286; HW04 Q1 (p. 17)

$$V_o \approx \begin{cases} V_{\text{sat}+}, & V_+ > V_- \\ V_{\text{sat}-}, & V_+ < V_- \end{cases}$$

Terms: V_+, V_-: non-inverting and inverting input voltages; $V_{\text{sat}+}, V_{\text{sat}-}$: positive and negative saturation outputs; V_o: comparator output. The homework assumes saturation at the supply rails. In its drawing, $V_- = V_1$ and $V_+ = V_2$. At equal inputs, this ideal switching rule does not determine the output.

F48 Input bias and input-offset currents

Slides p. 291; definitions expressed mathematically

$$I_B = \frac{I_{B+} + I_{B-}}{2}, \qquad I_{\text{OS}} = |I_{B+} - I_{B-}|$$

Terms: I_{B+}, I_{B-}: bias currents at the two op-amp inputs; I_B: average input bias current; I_{OS}: input-offset current magnitude. Use the manufacturer's current-direction convention when signs matter.

F49 Slew rate

Slides p. 292; definition expressed mathematically

$$\text{SR} = \max\left|\frac{dV_o}{dt}\right|, \qquad \Delta t \geq \frac{|\Delta V_o|}{\text{SR}}$$

Terms: SR: maximum output-voltage rate, often $\text{V}/\mu\text{s}$; V_o: output voltage; t: time; ΔV_o: required output change; Δt: minimum time for that change when slew-rate limited. The second form is a derived bound.

5 Filtering, bridges, grounding, power and PWM

F50 Optoisolator current transfer ratio

Slides p. 298

$$\text{CTR} = \frac{I_o}{I_i}, \qquad \text{CTR}_{\%} = 100\frac{I_o}{I_i}$$

Terms: I_i: input LED current; I_o: output phototransistor current; CTR: dimensionless current transfer ratio; $\text{CTR}_{\%}$: percentage form. For example, a percentage above 100% means the output current exceeds the input current.

F51 First-order RC cutoff and the half-power point

Slides pp. 303, 306; related to HW05 Q8 (p. 26)

$$f_c = \frac{1}{2\pi RC}, \qquad \frac{|V_o(f_c)|}{|V_{o,\text{pass}}|} = \frac{1}{\sqrt{2}} \approx 0.7071$$

Terms: f_c: cutoff frequency (Hz); R: filter resistance (Ω); C: capacitance (F); $V_o(f_c)$: output amplitude at cutoff; $V_{o,\text{pass}}$: reference passband output amplitude for the same input. The RC formula applies to the first-order circuit on p. 306; the amplitude ratio corresponds to half the reference power at the same impedance.

F52 Attenuation in decibels

Slides p. 304

$$A_{\mathrm{dB}} = 10\log_{10}\left(\frac{P_i}{P_o}\right) = 20\log_{10}\left|\frac{V_i}{V_o}\right| \quad \text{for equal resistances}$$

Terms: A_{dB}: positive loss/attenuation in dB; P_i, P_o: input and output powers; V_i, V_o: corresponding RMS voltage magnitudes. The voltage form requires equal real reference resistances; otherwise use the power ratio. At the half-power point, attenuation is about 3.01 dB, or gain is about −3.01 dB.

F53 Wheatstone bridge output

Slides pp. 310–312; HW05 Q6 (p. 25)

$$V_{AB} = V_s\frac{R_1}{R_1+R_2}, \qquad V_{AD} = V_s\frac{R_3}{R_3+R_4}$$

$$V_o = V_{AB} - V_{AD} = V_s\left(\frac{R_1}{R_1+R_2} - \frac{R_3}{R_3+R_4}\right)$$

Terms: $V_s = V_A - V_C$: excitation between supply nodes A, C; R_1, R_2: series path A–B–C; R_3, R_4: series path A–D–C; $V_{AB} = V_A - V_B$ and $V_{AD} = V_A - V_D$: upper-arm voltage drops; $V_o = V_D - V_B$: chosen output polarity. Assumes negligible current through the output measurement device.
Use: For a midpoint measured relative to ground, use the lower-arm voltage divider: $V_{\mathrm{mid}} = V_s R_{\mathrm{lower}}/(R_{\mathrm{upper}} + R_{\mathrm{lower}})$. This is F13 applied to a bridge leg; R_{upper} connects to supply and R_{lower} to ground. HW05 Q6 labels its nodes and resistors differently.

F54 Balanced Wheatstone bridge

Slides p. 311; HW05 Q9 (p. 27)

$$V_o = 0 \iff \frac{R_1}{R_2} = \frac{R_3}{R_4} \iff R_1R_4 = R_2R_3$$

Terms: R_1–R_4: bridge arms as defined in F53; V_o: differential bridge output. The two voltage-divider fractions must be equal at balance.
Use: In the *different labeling of HW05 Q9*, R_1, R_3 form the left leg and R_2, R_x the right leg. Therefore $R_1/R_3 = R_2/R_x$, or $R_3 = R_1R_x/R_2$, where R_x is the given right-lower resistor. These are the same balance law with renamed arms.

F55 Bridge output change from one changing resistor

Slides pp. 313–315; corrected small-change approximation

$$\Delta V_o = V_s\left[\frac{R_1+\Delta R_1}{R_1+\Delta R_1+R_2} - \frac{R_1}{R_1+R_2}\right]$$

$$\Delta V_o = \frac{V_sR_2\Delta R_1}{(R_1+R_2)(R_1+R_2+\Delta R_1)} \approx \frac{V_sR_2}{(R_1+R_2)^2}\Delta R_1$$

$$R_1 = R_2 = R: \qquad \Delta V_o \approx \frac{V_s}{4}\frac{\Delta R_1}{R}$$

Terms: V_s: bridge supply; R_1: changing sensor arm; R_2: fixed resistor in the same series leg; ΔR_1: sensor resistance change; ΔV_o: corresponding output change using F53 polarity; R: common nominal resistance in the equal-arm case. The approximation assumes $|\Delta R_1| \ll R_1 + R_2$. Other arms remain fixed and output loading is negligible.
Source note: The approximation printed on p. 313 omits the factor $R_2/(R_1 + R_2)$. Its equal-arm result, also used on p. 315, is twice the correct small-signal result. The approximation above follows directly by differentiating F53.

F56 RTD bridge temperature sensitivity and cold-junction matching

Slides pp. 327–328; corrected using F29 and F55

$$\Delta V_o \approx \frac{V_sR_2R_0\alpha}{(R_0+R_2)^2}\Delta T, \qquad K_B = \frac{V_sR_2R_0\alpha}{(R_0+R_2)^2}$$

$$K_B = k_{\mathrm{TC}} \quad \text{for local cold-junction compensation}$$

Terms: R_0: sensor-arm resistance at the reference temperature; R_2: fixed series arm; α: sensor temperature coefficient; V_s: excitation; ΔT: temperature change; ΔV_o: bridge-output change; K_B: bridge temperature sensitivity (V/K); k_{TC}: thermocouple sensitivity (V/K). Use polarity that adds the missing cold-junction EMF.
Source note: The slides propagate the approximation issue in F55 and omit the temperature factor in the displayed p. 328 EMF equation. K_B is a voltage *per degree*; multiplying it by ΔT gives a voltage.

F57 Differential voltage, common-mode voltage and CMRR

Slides p. 332

$$V_d = V_A - V_B, \qquad V_{\text{CM}} = \frac{V_A + V_B}{2}$$

$$\text{CMRR} = \left|\frac{A_d}{A_{\text{CM}}}\right|, \qquad \text{CMRR}_{\text{dB}} = 20\log_{10}\left|\frac{A_d}{A_{\text{CM}}}\right|$$

Terms: V_A, V_B: two input-line voltages relative to a common ground; V_d: differential voltage; V_{CM}: common-mode voltage; A_d: differential gain; A_{CM}: common-mode gain; CMRR: common-mode rejection ratio. Larger CMRR means less output contribution from a given common-mode input relative to differential gain.

F58 Load power and maximum power transfer

Slides pp. 340–342

$$I = \frac{E}{R+r}, \qquad P_L = I^2R = \frac{E^2R}{(R+r)^2}$$

$$\frac{dP_L}{dR} = \frac{E^2(r-R)}{(R+r)^3} = 0 \implies R = r, \qquad P_{\max} = \frac{E^2}{4r}$$

$$\text{AC conjugate match:} \qquad Z_L = Z_s^*$$

Terms: E: ideal source EMF (V); r: internal source resistance; R: load resistance; I: load current; P_L: load power (W); $P_{\max}$: maximum available power; Z_s: complex source impedance; Z_L: load impedance; *: complex conjugate. The DC maximum assumes $r > 0$.

Source note: The optimization variable is R, so p. 342 should use dP/dR, not dP/dt. For complex AC impedances, matching means conjugate matching. Maximum-power matching is a different design objective from minimizing voltage loading in F35.

F59 Pulse width modulation: duty cycle, period and average voltage

Slides p. 345; HW05 Q7 (p. 26)

$$D = \frac{t_{\text{on}}}{T}, \qquad D_{\%} = 100D, \qquad T = t_{\text{on}} + t_{\text{off}} = \frac{1}{f}$$

$$V_{\text{avg}} = DV_H + (1-D)V_L; \qquad V_L = 0 \implies V_{\text{avg}} = DV_H$$

Terms: D: duty fraction between 0 and 1; $D_{\%}$: percentage duty cycle; $t_{\text{on}}, t_{\text{off}}$: high and low durations; T: period; f: repetition frequency; V_H, V_L: high and low voltage levels; V_{avg}: time-average voltage over a cycle.

6 Digital conversion, sampling and data checks

F60 Positional number representation: decimal and binary

Slides pp. 351–360; HW05 Q3–Q4 (p. 24)

$$N = \sum_{i=0}^{n-1} d_i b^i$$

Terms: N: numerical value; b: base, 10 for decimal or 2 for binary; d_i: digit in position i; $i = 0$: rightmost/least-significant position; n: number of digits. For binary, each d_i is 0 or 1; adding the powers of 2 at the 1-bits converts a binary word to decimal. Finding those powers for a given N performs the reverse conversion.

F61 Digital levels and ADC resolution / LSB size

Slides pp. 169, 366–369; HW05 Q2 (p. 23), HW06 Q7 (p. 30)

$$N_L = 2^n, \qquad q = \frac{V_{\max} - V_{\min}}{2^n} = \frac{V_{\text{span}}}{2^n}, \qquad q_{\text{frac}} = \frac{1}{2^n}$$

Terms: n: converter word length in bits; N_L: number of possible codes/levels; $V_{\max}, V_{\min}$: input-range endpoints; V_{span}: full input span; q: one least-significant-bit (LSB) step in volts; q_{frac}: fractional full-scale resolution. Codes run from 0 to $2^n - 1$.

Source note: The 2^n quantity is the number of possible codes, not a number of bits. HW05 Q2 treats the stated 4 V as full-scale span; a single instantaneous input value alone would not determine resolution.

F62 Minimum ADC word length

Slides p. 370; HW05 Q1 (p. 23); derived by rearranging F61

$$2^n \geq \frac{V_{\text{span}}}{q_{\text{req}}}, \qquad n_{\min} = \left\lceil \log_2\left(\frac{V_{\text{span}}}{q_{\text{req}}}\right) \right\rceil$$

$$q_{\text{req}} = |K|\Delta x_{\text{req}}$$

Terms: $n_{\min}$: minimum whole number of bits; V_{span}: input-voltage span; q_{req}: largest acceptable voltage step; $\lceil\ \rceil$: round upward to the next integer; $\log_2$: base-2 logarithm; K: sensor sensitivity (V per unit input); Δx_{req}: required resolution of the measured physical quantity. Assumes a nondegenerate range and a uniform quantizer.

F63 Ideal quantization error

Slides p. 368; related to HW06 Q3 (p. 29)

$$e_q = V_q - V_i, \qquad -\frac{q}{2} \leq e_q \leq \frac{q}{2}$$

Terms: e_q: quantization error; V_i: actual sampled input; V_q: quantized representative voltage; q: LSB interval from F61. The $\pm\frac{1}{2}$ LSB bound assumes rounding/centered levels and no clipping or overload. It does not include sensor or converter nonlinearity errors.

F64 DAC output and the full-scale convention

Slides pp. 382–385

$$V_o = V_{\min} + D_c q_D$$

$$q_D = \frac{V_{\text{ref}}}{2^n} \text{ (reference-span convention)}, \qquad q_D = \frac{V_{\text{FS,code}} - V_{\min}}{2^n - 1} \text{ (endpoint convention)}$$

Terms: V_o: DAC output; $V_{\min}$: output for code zero; D_c: unsigned decimal input code, $0 \leq D_c \leq 2^n - 1$; q_D: volts per code step; n: bit count; V_{ref}: nominal reference span; $V_{\text{FS,code}}$: actual output at the all-ones code.
Source note: The example on p. 383 divides by 2^n while also calling the all-ones output exactly full scale. If the all-ones output must be exactly the stated voltage, use the $2^n - 1$ endpoint form. These two specifications should not be mixed.

F65 Sampling frequency, Nyquist frequency and sampling criterion

Slides pp. 371–379; HW06 Q4, Q8 (pp. 29–31)

$$f_s = \frac{1}{T_s}, \qquad f_N = \frac{f_s}{2}, \qquad f_s > 2f_{\max}$$

$$\text{Slide's practical rule of thumb:} \qquad f_s \approx 10 f_{\text{interest}}$$

Terms: f_s: sampling rate (samples/s or Hz); T_s: interval between samples (s); f_N: Nyquist frequency associated with that sampling rate; $f_{\max}$: highest signal frequency after anti-alias filtering; f_{interest}: signal frequency of interest for the slide's practical guideline.
Source note: The theoretical boundary is twice the highest frequency; use a strict margin to avoid the equality/phase problem and allow real filtering. The wording "Nyquist frequency + 1" on p. 375 and in HW06 Q8 mixes a frequency with a sample count and is not a general reconstruction formula. Keep the course wording in mind for that question, but use the dimensionally consistent relations above.

F66 Conversion-time limit on sampling

Slides p. 366

$$f_{s,\max} \leq \frac{1}{t_c}, \qquad f_{\text{signal}} < \frac{1}{2t_c}$$

Terms: t_c: ADC conversion time; $f_{s,\max}$: maximum sampling rate in a non-overlapped conversion model; f_{signal}: signal frequency to be sampled without aliasing in that model. The slide writes the limiting signal frequency as $1/(2t_c)$; practical acquisition and other overhead can lower it.

F67 Even and odd parity bits

Slides pp. 397–398, 433; HW06 Q5 (pp. 29–30)

$$p_{\text{even}} = b_0 \oplus b_1 \oplus \cdots \oplus b_{n-1}, \qquad p_{\text{odd}} = \overline{p_{\text{even}}}$$

Terms: b_i: data bit in position i; n: number of data bits; p_{even}: appended bit making the total number of 1s even; p_{odd}: appended bit making it odd; $\oplus$: exclusive OR (XOR); overbar: Boolean complement. A parity bit detects an odd number of bit flips, but an even number can go undetected.

F68 Additive checksum

Slides p. 399; arithmetic description expressed as a formula

$$C_{\text{sum}} = \left(\sum_{i=1}^{m} w_i \right) \bmod 2^k$$

Terms: w_i: numerical value of data word i; m: number of words; k: number of bits retained in the checksum; C_{sum}: resulting k-bit additive checksum; mod: remainder operation. If all sum bits are transmitted, omit the modulo operation.
Source note: The slide describes a binary sum, not a specific cryptographic hash or CRC. Its claim that every double error goes undetected is not generally true for all checksums.

7 Boolean logic formulas

F69 Basic logic gates

Slides pp. 408–423

Name	Formula	Output is 1 when
AND	$Q = A \cdot B$	Both inputs are 1.
OR	$Q = A + B$	At least one input is 1.
NOT	$Q = \overline{A}$	The input is 0.
NAND	$Q = \overline{A \cdot B}$	The inputs are not both 1.
NOR	$Q = \overline{A + B}$	Both inputs are 0.
XOR	$Q = A \oplus B = A\overline{B} + \overline{A}B$	The inputs differ.
XNOR	$Q = \overline{A \oplus B} = AB + \overline{A}\,\overline{B}$	The inputs are equal.

Terms: A, B: Boolean input bits; Q: Boolean output bit; $\cdot$ or juxtaposition: AND; $+$: OR, not numerical addition; overbar: NOT; $\oplus$: XOR. All variables are 0 or 1. XNOR appears in the overview on p. 408.
Source note: The body text on the XOR slide, p. 423, mistakenly says “NOR”; its equation and table describe XOR.

F70 De Morgan forms and gates built from NAND/NOR

Slides pp. 426–427; formulas expressed from the circuits

$$\overline{\overline{A} + \overline{B}} = AB, \qquad \overline{\overline{A}\,\overline{B}} = A + B$$
$$\overline{A + A} = \overline{A}, \qquad \overline{AA} = \overline{A}$$

Terms: A, B: Boolean inputs; overbar: NOT; $+$: OR; multiplication: AND. The first expression is the three-NOR implementation of AND; the second is the three-NAND implementation of OR. The last two expressions show how tied inputs make an inverter.

F71 Examples of combined logic

Slides pp. 407, 425, 433; expressions derived from the stated rules and diagrams

$$Q_{\text{practice}} = A\overline{B}, \qquad Q_{\text{buzzer}} = KD + HD_d$$
$$L_R = \overline{A}\,\overline{B}, \qquad L_A = \overline{A}B, \qquad L_G = A\overline{B}$$

Terms: Q_{practice}: output of the AND-with-inverted-B example; A, B: input bits, also used as the two traffic-light code bits in the last line; Q_{buzzer}: car buzzer output; K: key-in-ignition signal; D: a-door-open signal; H: headlights-on signal; D_d: driver’s-door-open signal; L_R, L_A, L_G: red, amber and green lamp commands in the p. 433 circuit. A value of 1 means the condition is true or the output is on.

8 Programming-example calculation formulas

F72 Sum, accumulation and the homework loop

Slides pp. 104–105, 116, 140–142; HW02 Q4 (p. 8)

$$S = \sum_{i=1}^{m} x_i, \qquad S_{j+1} = S_j + x_{j+1}$$

$$\text{HW02 loop:} \qquad S = \sum_{i=0}^{n-1} i = \frac{n(n-1)}{2}$$

Terms: S: total; x_i: value i in a list of m values; S_j: accumulated sum after j values; n: positive integer input to the homework loop; i: loop counter starting at 0. The loop uses $i < n$, so n is excluded. In code, the update `sum = sum + i` assigns a new value; it is not an algebraic equality between an unchanged variable and itself plus i.

F73 Arithmetic mean / average-temperature example

Slides pp. 116, 143–145

$$\overline{x} = \frac{1}{m}\sum_{i=1}^{m} x_i, \qquad T_{\text{avg}} = \frac{T_{\max} + T_{\min}}{2}$$

Terms: $\overline{x}$: mean of m values x_i; $T_{\max}, T_{\min}$: daily maximum and minimum temperatures; T_{avg}: their arithmetic average, as defined in the example. That midpoint need not equal a time-weighted daily average.

F74 Fahrenheit-to-Celsius conversion

Slides p. 116

$$T_C = \frac{5}{9}(T_F - 32)$$

Terms: T_C: numerical temperature in degrees Celsius; T_F: numerical temperature in degrees Fahrenheit; 32: Fahrenheit value of 0 °C; 5/9: ratio of Celsius to Fahrenheit temperature intervals.

F75 Percentage tax and total price

Slides pp. 116–117

$$P_{\text{tax}} = r_{\text{tax}} P_0, \qquad P_{\text{total}} = P_0 + P_{\text{tax}} = P_0(1 + r_{\text{tax}})$$

Terms: P_0: original price; r_{tax}: tax rate as a fraction (0.10 in the slide's example); P_{tax}: tax amount; P_{total}: price including tax.

F76 Rectangle area and mowing time

Slides pp. 146–147

$$A_{\text{lot}} = L_{\text{lot}} W_{\text{lot}}, \qquad A_{\text{house}} = L_{\text{house}} W_{\text{house}}$$

$$A_{\text{mow}} = A_{\text{lot}} - A_{\text{house}}, \qquad t_{\text{mow}} = \frac{A_{\text{mow}}}{r_{\text{mow}}}$$

Terms: L, W: length and width of the indicated rectangular lot or house; $A_{\text{lot}}, A_{\text{house}}$: their areas; A_{mow}: grass area to mow; r_{mow}: area mowed per unit time; t_{mow}: mowing time. With area in m^2 and rate in m^2/min, time is in minutes.

F77 Gross pay with overtime

Slides pp. 87–90, 148

$$P = \begin{cases} hr, & 0 \le h \le 40 \\ 40r + (h-40)r_o, & h > 40 \end{cases}$$

Terms: P: gross pay; h: hours worked; r: regular hourly pay rate; r_o: overtime hourly pay rate. The example applies regular pay to the first 40 hours; r_o is a rate, not necessarily a multiplier. Without overtime the formula is $P = hr$.

F78 Minimum of three values

Slides p. 150; mathematical expression of the programming task

$$x_{\min} = \min(x_1, x_2, x_3)$$

Terms: x_1, x_2, x_3: the three input numbers; $x_{\min}$: smallest input; min: minimum-selection operation. Comparisons in a flowchart implement this operation.

9 Electromagnetism equations shown in the slides

F79 Maxwell's equations in vacuum: differential form

Slides p. 335; all four equations appear in the slide image

$$\nabla \cdot \mathbf{E} = \frac{\rho_e}{\varepsilon_0} \qquad \text{Gauss's law for electricity}$$

$$\nabla \cdot \mathbf{B} = 0 \qquad \text{Gauss's law for magnetism}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \text{Faraday's law of induction}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} \qquad \text{Ampère–Maxwell law}$$

Terms: $\mathbf{E}$: electric field (V/m); $\mathbf{B}$: magnetic flux density (T); ρ_e: electric charge density (C/m^3), called ρ in the slide; ε_0: vacuum permittivity (F/m); μ_0: vacuum permeability (H/m); $\mathbf{J}$: electric current density (A/m^2), called $\mathbf{j}$ in the slide; c: speed of light in vacuum; t: time; $\nabla\cdot$: divergence, measuring local outward flux; $\nabla\times$: curl, measuring local circulation; $\partial/\partial t$: partial time derivative. Here $1/c^2 = \mu_0\varepsilon_0$. These equations provide the electromagnetic context for induced voltages and interference.

10 Homework question-to-formula lookup

The entries below identify the formula to use without repeating the formula catalog. Page references in this table refer to *HW_fall_2026.pdf*. Pure terminology, component-identification, and other concept-only questions do not need an additional formula.

Homework	PDF p.	Required relationship	Entry
HW01 Q1	1	Set point minus measured temperature	F01
HW02 Q4	8	Sum of loop-counter values from 0 through $n-1$	F72
HW03 Q5	14–15	Percentage of full-scale span	F04
HW03 Q6	15	Strain from resistance change and gauge factor	F15
HW03 Q7	15	Encoder angle per slot/count	F18
HW03 Q9	16	Tracking-wheel travel converted to pulses	F19
HW03 Q10	16	Measured value minus true value	F03
HW04 Q1	17	Comparator input ordering and saturation	F47
HW04 Q5	18–19	Unity-gain voltage follower (equation-based recognition)	F39
HW04 Q6	19–20	Difference-amplifier gain from resistor ratios	F42
HW04 Q9	21	Fraction of temperature change after one time constant	F11
HW04 Q10	21–22	Inverting-amplifier output voltage	F40
HW05 Q1	23	Minimum bits for a specified voltage resolution	F62
HW05 Q2	23	Full-scale span divided by number of ADC codes	F61
HW05 Q3	24	Decimal-to-binary positional representation	F60
HW05 Q4	24	Binary-to-decimal weighted sum	F60
HW05 Q6	25	Bridge midpoint voltage-divider behavior	F53
HW05 Q7	26	PWM high-time divided by complete period	F59
HW05 Q9	27	Bridge balance; use the homework's resistor labels	F54
HW06 Q5	29–30	Even-parity bit from the XOR of data bits	F67
HW06 Q7	30	ADC voltage resolution	F61
HW06 Q8	30–31	Sampling criterion; see the source-wording correction	F65

Related conceptual questions. HW04 Q2 concerns ideal-op-amp properties (F37). HW05 Q8 concerns the cutoff definition (F51). HW06 Q3 concerns the origin of quantization error (F63), and HW06 Q4 concerns Nyquist terminology (F65). These do not add distinct calculation formulas.

Scope of deduplication. A repeated slide, a numerical worked example, a change of variable names, or an algebraic rearrangement does not create a new formula entry. Different circuit configurations, sign conventions, and converter full-scale conventions are retained because they change how an equation is used. Generic assignment syntax and arbitrary demonstration values in pseudocode are not separate physical or calculation formulas.